

AMELIA NGUYEN

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EDUCATION

UNIVERSITY OF MICHIGAN - DEARBORN

Dearborn, MI

Master of Science in Data Science, GPA: 4.0.

Expected Graduation: May 2027

Relevant Coursework: Machine Learning and Computational Statistics, Regression Analysis, Database Systems, Big Data, Operations Research, Data Analytics & Modeling. *Planned:* Deep Learning, Advanced Artificial Intelligence, Optimization

FOREIGN TRADE UNIVERSITY

Hanoi

Bachelor of International Business, GPA: 3.8.

May 2023

TECHNICAL SKILLS

Machine Learning & AI: Supervised Learning (Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, SVM, Naive Bayes, KNN), Unsupervised Learning (K-Means, Clustering, PCA), Deep Learning (CNNs, RNNs, LSTM, Transformer), Reinforcement Learning, Recommendation Systems, Time Series, Feature Engineering, Model Evaluation, Hyperparameter Optimization, Computer Vision, LLMs, GenAI, Agentic AI, Hugging Face

Libraries: Python (Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn, TensorFlow, PyTorch), SQL, PySpark

Cloud & MLOps: Google Cloud Platform (BigQuery, Vertex AI, Cloud Storage), Spark, Hadoop

Visualization & Analytics: Tableau, Power BI, Google Analytics, Excel, SAP HANA, SAP Analytics Cloud

Automotive: Kalman Filtering, Robotics, Telematics, Autonomous Driving, State Estimation, Localization, Control Systems

PROFESSIONAL EXPERIENCE

UNIVERSITY OF MICHIGAN

Dearborn, MI

Research Assistant

Jun 2026 - Present

- Conducted Computer Vision research experiments in emotion detection and secure biometrics at the Trustworthy AI Lab.

GEARLAUNCH

Hanoi

Data Analyst

Nov 2023 - Jul 2025

- Served as the **sole data analyst**, managing the company's end-to-end internal analytics infrastructure and presenting business analysis insights directly to C-level leadership and department heads.
- Designed and deployed Machine Learning (Regression, Random Forest, XGBoost) and Time Series Forecasting (AR, ARIMA, SARIMAX) to support e-commerce executive strategy development, resulting in a 20% increase in annual sales KPI.
- Implemented and optimized scalable data pipelines using SQL, Python, Google Cloud Platform (Cloud Run, VertexAI), and Tableau, streamlining reporting workflows and increasing data processing efficiency by 3 times without additional infrastructure.
- Analyzed data from 35,000 monthly active user stores, resulting in an average of a 50% increase in marketing conversions.

VCCORP CORPORATION

Hanoi

Data Analyst - Team Lead

May 2023 - Nov 2023

- Managed a team of four to perform full-cycle A/B testing (PySpark, Python, SQL, HDFS, ElasticSearch) for recommendation engine deployment on 7 digital media websites, resulting in 10–30% monthly gains in core web KPIs.
- Designed and built an automated A/B testing evaluation framework for the internal platform and a client-facing product.
- Maintained primary responsibility for advanced statistical analysis for 3 digital publishing platforms for product teams.

Data Analyst

March 2022 - May 2023

- Designed and monitored 30+ KPIs with automated dashboards and provided ad hoc analysis for 5 digital publishing products.
- Developed user segmentation analysis that increased engagement with the app notification system by 30% within 3 months.

PROJECTS

Street Obstacle Detection | PyTorch, Computer Vision, Hugging Face

June 2026

Deployed apps for street obstacle detection using fine-tuned YOLOv11n and RT-DETR transformer-based models.

Fuel Consumption Prediction for ICE powertrain vehicles | Python, XGBoost, LightGBM, LSTM, TCN

Apr 2026 - May 2026

Achieved a 13.87% MAPE and 0.3 MAE on real-world ODB-II telematics data collected from 300 vehicles throughout a year.

Deer-Vehicle Collision Prediction | Python, Random Forest, Gradient Boosting, K-Means

Apr 2026

Predicted deer collision risk with supervised methods and K-Means clustering with PCA.

Vehicle Crash Frequency and Severity Prediction | Python, Regression, Classification

Nov 2025 - Dec 2025

Deployed predictive modeling and diagnostics using linear regression and logistic regression with 85% accuracy.

CERTIFICATIONS

Self-Driving Cars Specialization (University of Toronto Online) | Simulation, Control System, AI

Jul 2026 - Present

AI for Autonomous Vehicles and Robotics (University of Michigan Online) | Robotics, Autonomous Driving Systems

Jul 2026